

Notes: Easley, Hvidkjaer, and O’Hara (JF, 2002)

Is Information Risk a Determinant of Asset Returns?

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1 One-sentence summary

This note summarizes Easley, Hvidkjaer, and O'Hara (2002), which asks whether private-information risk is priced in the cross-section of stock returns, estimates each NYSE stock's probability of information-based trading (PIN) from high-frequency buy-sell order-flow data using a structural market microstructure model, and shows in Fama-MacBeth asset-pricing tests that stocks with higher PIN earn significantly higher expected returns even after controlling for beta, size, book-to-market, spreads, return volatility, and turnover.

2 Big picture

The paper connects market microstructure to asset pricing. Standard asset-pricing models usually treat prices as if information is already incorporated, while microstructure models study how information enters prices through trading. The authors argue that this distinction matters: if price discovery is a dynamic process, then the risk of trading against better-informed investors can affect required returns.

The empirical object is the probability of information-based trading, denoted *PIN*. The paper estimates *PIN* for individual NYSE-listed stocks from 1983 to 1998 and asks whether higher *PIN* predicts higher cross-sectional returns.

The central result is economically large: a 10 percentage-point difference in *PIN* corresponds to about 2.5 percent per year higher expected returns. This suggests that information asymmetry is not only a trading-cost or spread issue; it is a priced risk characteristic.

3 Research question

The paper asks three closely related questions.

1. **Can private information affect equilibrium required returns?** The theory section provides a rational-expectations example in which private information makes uninformed traders bear risk they cannot fully diversify away.
2. **Can the amount of private-information trading be measured from data?** The paper uses a structural sequential-trade model to estimate *PIN* from the daily number of buys and sells.
3. **Is information risk priced in the cross-section?** The paper adds *PIN* to the Fama and French (1992) cross-sectional return framework.

4 Incremental contribution

4.1 Conceptual contribution

The paper's conceptual contribution is to move information asymmetry from a microstructure friction to an asset-pricing variable. Earlier microstructure work showed that private information

affects bid-ask spreads and price formation. This paper asks whether private-information risk also affects expected returns.

The key idea is that private information changes who holds what risk. If informed traders know more about which stocks are attractive, uninformed traders cannot simply diversify the disadvantage away. They may hold too much of bad-information stocks and too little of good-information stocks. The required return must compensate them for this informational disadvantage.

4.2 Measurement contribution

The paper uses a structural trade model to construct a stock-level proxy for information risk. This is important because private information is not directly observable. The *PIN* measure is derived from the frequency of abnormal buy or sell imbalances relative to normal uninformed trading intensity.

Unlike bid-ask spreads or Amihud-style price impact measures, *PIN* is not constructed from returns or prices. It is based on trade direction and order arrivals, which helps avoid the critique that the variable is simply a noisy transformation of price or beta.

4.3 Empirical contribution

The paper estimates *PIN* for a large panel of NYSE stocks over 1983-1998 and incorporates it into monthly cross-sectional return regressions. The main empirical contribution is that *PIN* remains positive and significant after controlling for the usual Fama-French variables and after adding spreads, return volatility, turnover, and turnover volatility.

This makes the paper one of the first large-scale empirical demonstrations that market microstructure-based information risk is priced in expected returns.

5 Theory intuition

5.1 Static efficiency versus dynamic price discovery

Traditional asset-pricing models often assume that if markets are efficient, information is already in prices. The authors argue that this view is incomplete because efficiency is not a static state; it is a process. Prices become efficient through trading, and informed trading changes the risk borne by uninformed investors while prices adjust.

5.2 Heterogeneous beliefs and private information

The paper begins with a simple two-state, two-asset example. When all traders have common beliefs, they hold the same risk-free portfolio and there is no compensation for idiosyncratic risk. When traders have different beliefs, they choose different portfolios because each trader perceives different assets as mispriced.

The authors then replace arbitrary belief disagreement with private information. Traders share common priors, but one trader observes a signal that is not fully revealed by prices because noisy

supply prevents full revelation. In equilibrium, informed and uninformed traders hold different portfolios. The uninformed trader rationally faces information risk.

5.3 Private versus public information

The paper's example compares two economies. In one, information is private and prices do not fully reveal it. In the other, the same information is public. When information is public, traders agree on beliefs and hold common portfolios. When information is private, uninformed traders require higher expected returns because they bear the risk of trading against better-informed investors.

The main theoretical message is therefore: the same information can have different asset-pricing implications depending on whether it is public or private.

6 Microstructure model and construction of PIN

6.1 Sequential-trade environment

The paper uses a sequential trade model. At the beginning of each trading day, nature determines whether an information event occurs. If no event occurs, only uninformed buy and sell orders arrive. If an event occurs, informed traders also arrive and trade in the direction of the signal.

The model has five key parameters:

- α : probability that an information event occurs on a trading day.
- δ : probability that the information event is bad news.
- μ : arrival rate of informed orders on information-event days.
- ϵ_b : arrival rate of uninformed buy orders.
- ϵ_s : arrival rate of uninformed sell orders.

6.2 Likelihood function

For each stock-day, the data are the number of buyer-initiated trades B and seller-initiated trades S . The likelihood is a mixture of three cases: no news, bad news, and good news. On bad-news days, sell arrivals include informed sellers; on good-news days, buy arrivals include informed buyers.

The paper maximizes the likelihood stock by stock and year by year to estimate the five parameters. The structure of abnormal buy/sell imbalance identifies the informed-trading component, while normal buy and sell counts identify uninformed trading intensities.

6.3 PIN formula

The probability that a trade is information based is

$$PIN = \frac{\alpha\mu}{\alpha\mu + \epsilon_s + \epsilon_b}.$$

The numerator is the arrival rate of informed orders. The denominator is the total arrival rate of all orders. A higher *PIN* means a larger fraction of trades is expected to come from informed traders.

7 Data and sample

The sample consists of ordinary common stocks listed on the NYSE from 1983 to 1998. The authors exclude REITs, foreign-incorporated firms, closed-end funds, and stock-years with too few days of quotes or trades. The final annual sample ranges from 1,311 to 1,846 stocks.

The authors use ISSM and TAQ transactions data to classify trades as buys or sells using the Lee-Ready algorithm. Asset-pricing tests require CRSP and Compustat data, so the return-test sample is the intersection of NYSE stocks with available CRSP/Compustat variables. The monthly return regressions run from 1984 to 1998 because lagged *PIN* is used.

8 Empirical design

8.1 Portfolio sorts

The paper first sorts stocks independently by *PIN* and size. The basic prediction is that, within size groups, high-*PIN* stocks should earn higher returns. This portfolio exercise provides a transparent first look before formal regressions.

8.2 Fama-MacBeth asset-pricing regressions

The main regression follows Fama and French (1992). For each month, the authors estimate

$$R_{it} = \gamma_{0t} + \gamma_{1t}\beta_p + \gamma_{2t}PIN_{i,t-1} + \gamma_{3t}SIZE_{i,t-1} + \gamma_{4t}BM_{i,t-1} + \eta_{it}.$$

- R_{it} is the monthly excess return.
- β_p is the post-ranking portfolio beta.
- $PIN_{i,t-1}$ is the lagged annual probability of information-based trading.
- $SIZE$ is log market equity.
- BM is log book-to-market.

The coefficients are averaged over time using standard Fama-MacBeth methods. The paper also reports Litzenberger-Ramaswamy precision-weighted averages.

8.3 Errors-in-variables check

Because *PIN* is estimated, it may suffer from measurement error. The authors construct portfolio-level *PIN*, denoted *PPIN*, in the same spirit as beta portfolios. The evidence suggests that using *PPIN* does not materially alter the estimated effect and mostly adds noise rather than correcting a major measurement-error problem.

8.4 Alternative explanations

The paper tests whether *PIN* merely proxies for:

- bid-ask spread;
- daily return volatility;
- share turnover;
- turnover volatility.

Across these tests, *PIN* remains positive and significant. This is the main robustness result.

9 Main findings

9.1 *PIN* is stable and economically meaningful

The estimated *PIN* has a mean around 0.19 and a median around 0.18. Cross-sectionally, it is negatively correlated with trading volume and firm size: larger and more actively traded stocks tend to have lower information-based trading probabilities.

The annual *PIN* estimates are also stable over time. This supports the interpretation that *PIN* captures a stock's underlying information environment rather than transient trading noise.

9.2 High-*PIN* stocks earn higher returns

Portfolio sorts show that returns generally increase as one moves from low to high *PIN* within size groups. In formal Fama-MacBeth regressions, the coefficient on *PIN* is positive and statistically significant.

Using the precision-weighted specification, the coefficient on *PIN* is 2.086. This means that a 10 percentage-point increase in *PIN* implies about 0.21 percent higher monthly return, or roughly 2.5 percent per year.

9.3 *PIN* is not just spread, volatility, or turnover

The paper shows that spreads, return volatility, turnover, and turnover volatility cannot explain away the *PIN* effect. Even when all these variables are included jointly, *PIN* remains positive and statistically significant.

This is central because it shows that private-information risk is not simply a re-labeled liquidity measure.

10 Identification and interpretation

The paper is not a randomized identification design. It is best read as a structural measurement exercise plus cross-sectional asset-pricing evidence. The strongest interpretation is that stocks

with higher estimated private-information trading risk have higher expected returns, conditional on standard asset-pricing variables and other trading-friction controls.

A key advantage is that PIN is estimated from the trading process rather than constructed from prices. This makes the result less vulnerable to the critique that return regressions are mechanically driven by price-based variables. A limitation is that PIN is itself model-based: if the sequential-trade model mismeasures information risk, the coefficient may reflect model-specific features of trading rather than the true latent information environment.

11 Tables and figures

11.1 Table I: Equilibrium Prices and Shadow Risk-free Rates

11.2 Figure 1: Tree diagram of the trading process

Table I
Equilibrium Prices and Shadow Risk-free Rates

Date t State	Price of Asset 1	Price of Asset 2	Risk-free Rate
z_1	$5/2\rho$	$1/2\rho$	ρ^{-1}
z_2	$5/4\rho$	$5/4\rho$	$6/5\rho^{-1}$
z_3	$5/4\rho$	$5/4\rho$	$6/5\rho^{-1}$
z_4	$1/2\rho$	$5/2\rho$	ρ^{-1}

Table I: Equilibrium Prices and Shadow Risk-free Rates

Read: Table I shows equilibrium prices and shadow risk-free rates in the rational-expectations example with noisy supply. Prices are fully revealing in some states and nonrevealing in others.

Economic message: The table demonstrates how private information creates heterogeneous beliefs and affects expected excess returns, even in a small economy.

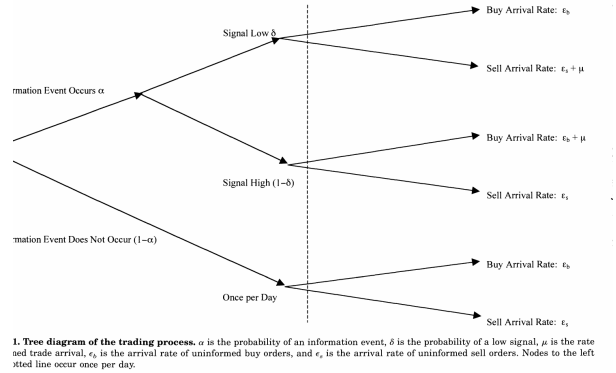


Figure 1: Tree diagram of the trading process

Read: Figure 1 lays out the sequential-trade structure. An information event may occur, the signal may be good or bad, and the resulting order arrival rates differ for buys and sells.

Economic message: The figure is the measurement foundation of the paper. It shows how abnormal order imbalances identify informed trading.

11.3 Figure 2: Parameter distributions - Panels A and B

11.4 Figure 2 Continued: Parameter distributions - Panel C

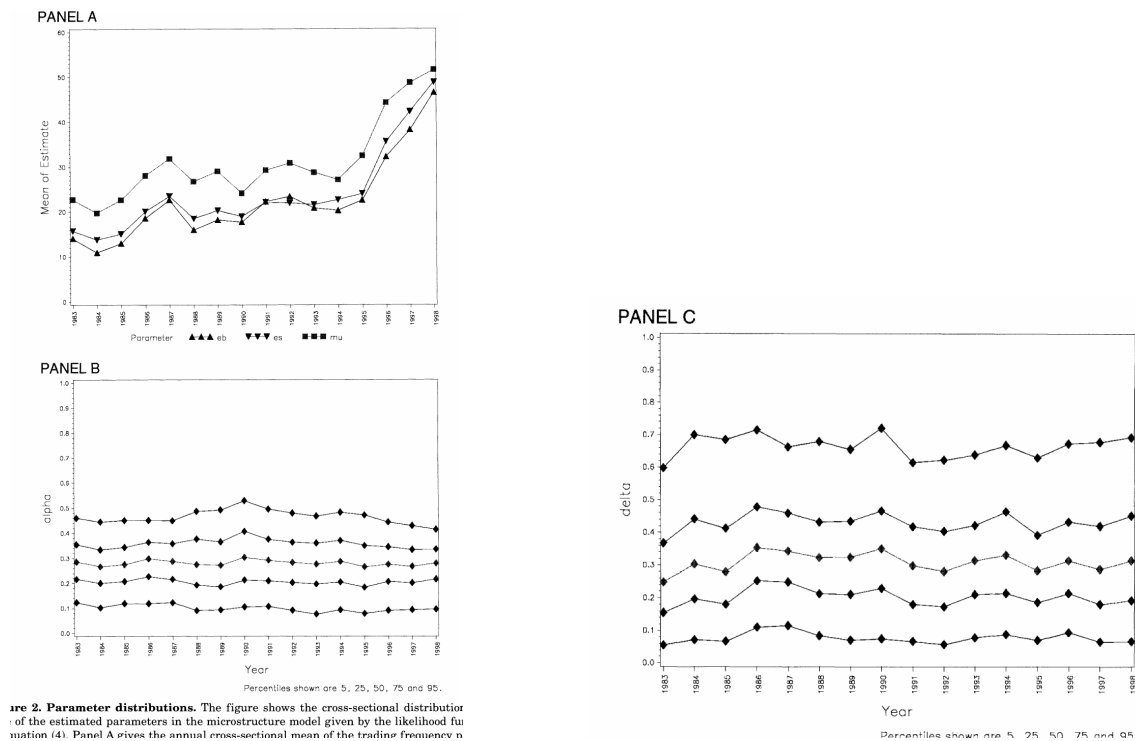


Figure 2: Parameter distributions - Panels A and B

Read: Panels A and B report the time-series behavior of the trading-frequency parameters and the distribution of α . The trading frequency parameters rise over time, while α is much more stable.

Economic message: Order flow grows as NYSE trading becomes more active, but the probability of information events remains a more persistent stock-level information characteristic.

Figure 2 Continued: Parameter distributions - Panel C

Read: Panel C reports the cross-sectional distribution of δ , the probability that an information event is bad news. The distribution is dispersed but stable across years.

Economic message: The bad-news probability is an information-event feature rather than a trading-volume feature.

11.5 Figure 3: PIN distributions

11.6 Figure 4: Parameter distributions with pooled data - Panels A and B

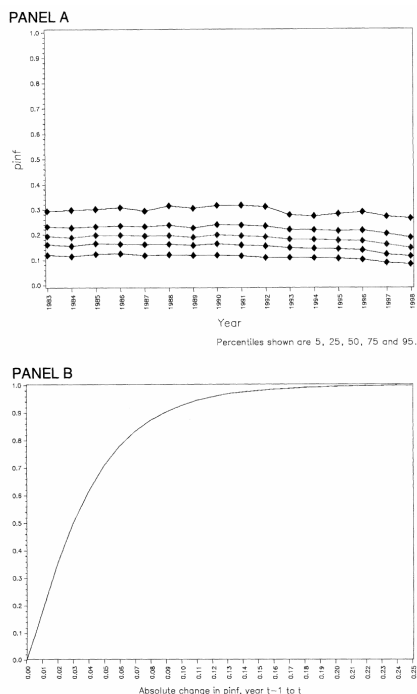


Figure 3. PIN distributions. The figure shows the cross-sectional distribution of the probability of information based trading, PIN , given by equation (5). Panel A shows

Figure 3: PIN distributions

Read: Figure 3 shows that the cross-sectional distribution of PIN is stable, with the median around 0.19. The cumulative distribution of absolute changes shows that most year-to-year changes are small.

Economic message: PIN behaves like a persistent information-environment characteristic, not a short-lived microstructure artifact.

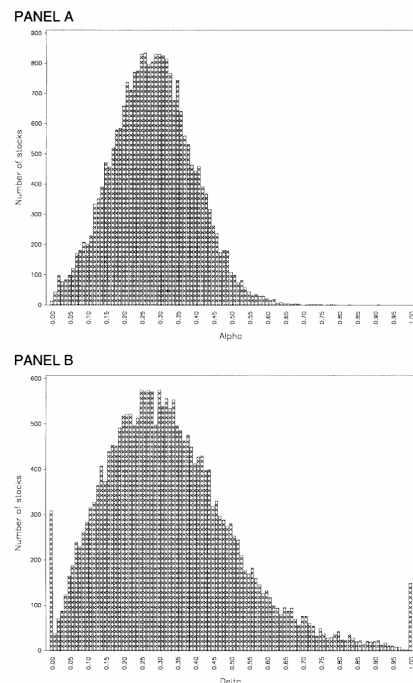


Figure 4. Parameter distributions with pooled data. The figure shows the empirical distribution of the microstructure model parameters with all stocks and all years pooled. Panel A shows the distribution of α , the probability that an information event has occurred.

Figure 4: Parameter distributions with pooled data - Panels A and B

Read: Figure 4 Panels A and B pool all stocks and years to show the distributions of α and δ . α is roughly bell-shaped, while δ is more skewed.

Economic message: The primitive information parameters vary across stocks, helping generate cross-sectional variation in PIN .

11.7 Figure 4 Continued: Parameter distributions with pooled data - Panels C and D

11.8 Table II: Parameter Summary Statistics

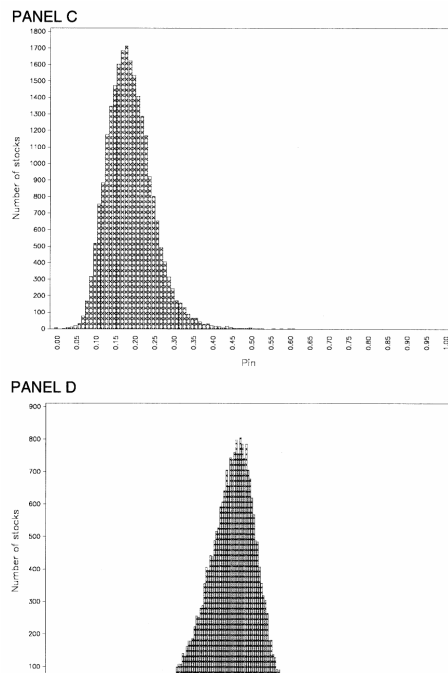


Figure 4 Continued: Parameter distributions with pooled data - Panels C and D

Read: Panel C shows the pooled distribution of PIN , concentrated around 0.18. Panel D shows the uninformed order-flow imbalance, centered close to zero.

Economic message: The distribution of PIN is tight but not degenerate, giving meaningful cross-sectional variation for asset-pricing tests.

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Table II
Parameter Summary Statistics

The table contains time-series averages across years 1983 to 1998 of cross-sectional means, medians, standard deviations, and the median of parameter standard errors from the likelihood estimation. The following are variable descriptions: α is the probability of an information event, δ is the probability of a low signal, μ is the rate of informed trade arrival, ϵ_b is the arrival rate of uninformed buy orders, and ϵ_s is the arrival rate of uninformed sell orders. PIN is the probability of informed trading as given by equation (5).

Variable	Mean	Median	Std. Dev.	Median Std. Err
α	0.283	0.281	0.111	0.035
δ	0.331	0.309	0.181	0.066
PIN	0.191	0.185	0.075	0.006

Table II: Parameter Summary Statistics

Read: Table II reports mean and median estimates of the model parameters. Mean PIN is 0.191 and median PIN is 0.185.

Economic message: Roughly one in five trades is estimated to be information based for the typical stock-year.

11.9 Table III: Portfolio Excess Returns

11.10 Table IV: Summary Statistics

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Table III
Portfolio Excess Returns

The table contains results for portfolios of stocks sorted independently by size and PIN, where size is the market value of equity measured at the end of year $t - 1$ and PIN is the probability of informed trading as given by equation (5) estimated over year $t - 1$. For each year, stocks are sorted into three PIN groups, ranging from low to high, and five size groups, ranging from small to large. The reported results are averages of the relevant variables over the sample period 1984 to 1998. Panel A reports average excess returns, Panel B reports the average number of stocks in each portfolio, Panel C reports the average size in \$ millions of the stocks in each portfolio, and Panel D reports the average PIN of stocks in each portfolio.

Size/PIN	Low	Medium	High	Size/PIN	Low	Medium	High
Panel A: Excess Returns				Panel B: Number of Stocks			
Small	0.148	0.202	0.474	Small	21.9	60.1	142.
2	0.462	0.556	0.743	2	30.5	79.0	115.
3	0.647	0.695	0.892	3	54.4	95.5	75.
4	0.873	0.837	0.908	4	93.4	99.0	39.

Table III: Portfolio Excess Returns

Read: Table III sorts stocks independently by size and PIN. Within most size groups, excess returns rise from low-PIN to high-PIN portfolios.

Economic message: The raw portfolio evidence already points to a positive relationship between information risk and expected returns.

11.11 Table V: Simple Correlations

11.12 Table VI: Asset-Pricing Tests

Table V
Simple Correlations

The table contains the time-series means of monthly bivariate correlations of the variables in the asset-pricing tests. RETURN is the percentage monthly return in excess of the one-month T-bill rate. BETAs are portfolio betas estimated from the full period using 40 portfolios. PIN is the probability of informed trading given by equation (5) and estimated yearly for each stock. PPIN is the portfolio PIN calculated by first sorting all stocks into 40 portfolios according to PIN each year, and then taking the average within each portfolio of the individual stock PIN in the following year. SIZE is the logarithm of year-end market value of equity. BM is the logarithm of book value of equity divided by market value of equity. SPREAD is the yearly average of the daily opening spreads of stock i . STD is the daily return standard deviation for stock i in year t . TURNOVER is the logarithm of the average monthly turnover year $t - 3$ to $t - 1$, and CVTURN is the logarithm of the coefficient of variation of the monthly turnover year $t - 3$ to $t - 1$.

Table V: Simple Correlations

Read: Table V shows that PIN is strongly negatively correlated with size and positively correlated with spreads, return volatility, and turnover volatility.

Economic message: High-PIN stocks are generally smaller and less liquid, so the formal regressions must separate information risk from standard liquidity and size effects.

Table IV
Summary Statistics

The table contains means, medians, minimum value, and maximum values on the variables included in the asset-pricing tests. All statistics are calculated from the full sample, that is, pooling 18 months. RETURN is the percentage monthly return in excess of the one-month T-bill rate. BETAs are portfolio betas estimated from the full period using 40 portfolios. PIN is the probability of informed trading given by equation (5) and estimated yearly for each stock. PPIN is the portfolio PIN calculated by first sorting all stocks into 40 portfolios according to PIN each year and then taking the average within each portfolio of the individual stock PIN in the following year. SIZE is the logarithm of year-end market value of equity. BM is the logarithm of book value of equity divided by market value of equity. SPREAD is the yearly average of the daily opening spreads of stock i . STD is the daily return standard deviation for stock i in year t . TURNOVER is the logarithm of the average monthly turnover year $t - 3$ to $t - 1$, and CVTURN is the logarithm of the coefficient of variation of the monthly turnover year $t - 3$ to $t - 1$.

Table IV: Summary Statistics

Read: Table IV reports summary statistics for return-test variables. The mean monthly excess return is 0.73 percent and mean PIN is 0.19.

Economic message: The test sample has meaningful variation in beta, size, book-to-market, spread, volatility, turnover, and PIN.

Table VI
Asset-Pricing Tests

The table contains time-series averages of the coefficients in cross-sectional asset-pricing regressions using the standard Fama and MacBeth (1973) methodology and Litzenger and Ramo (1979) precision-weighted means (weighted least-squares (WLS)). The dependent variable is the percentage monthly return in excess of the one-month T-bill rate. BETAs are portfolio betas estimated from the full period using 40 portfolios. PIN is the probability of informed trading in stock i of year $t - 1$. SIZE is the logarithm of market value of equity of stock i at the end of year $t - 1$, and BM is the logarithm of the ratio of book value of equity to market value of equity of stock i at the end of year $t - 1$.

Table VI: Asset-Pricing Tests

Read: Table VI reports the baseline Fama-MacBeth regressions. In the precision-weighted specification, the coefficient on PIN is 2.086 with a t-statistic of 4.362.

Economic message: Information-based trading risk is priced after controlling for beta, size, and book-to-market.

11.13 Table VII: Errors in Variables

11.14 Table VIII: Alternative Explanations

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Table VII
Errors in Variables

The table contains time-series averages of the coefficients in cross-sectional asset-pricing tests using standard Fama and MacBeth (1973) methodology and Litzenberger and Ramaswamy (L-R, 1979) precision-weighted means. The dependent variable is the percentage monthly return in excess of the one-month T-bill rate. Betas are portfolio betas calculated from the full period using 40 portfolios. *PIN* is the probability of information-based trading in stock *i* of year *t* - 1. *PPIN* is the portfolio pin calculated by first sorting all stocks into 40 portfolios according to *PIN* each year, and then taking the average within each portfolio of the individual stock *PIN* in the following year. *SIZE* is the logarithm of market value of equity (*ME*) in firm *i* at the end of year *t* - 1, and *BM* is the logarithm of the ratio of book value of common equity to market value of equity for firm *i* in year *t* - 1. *T*-values are given in parentheses.

	<i>Beta</i>	<i>PIN</i>	<i>PPIN</i>	<i>SIZE</i>	<i>BM</i>
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Table VII: Errors in Variables

Read: Table VII replaces individual *PIN* with portfolio *PIN*, *PPIN*, and also includes both variables. The original *PIN* remains the more informative measure.

Economic message: Measurement error in *PIN* does not appear to drive the main result.

12 Short summary

The paper estimates the probability of information-based trading from transaction data and shows that this information-risk measure predicts expected returns. The logic is that uninformed traders face a systematic disadvantage when information is private and not fully revealed by prices. Empirically, high-*PIN* stocks earn higher returns even after standard risk, size, value, spread, volatility, and turnover controls.

13 Conclusion

13.1 Core conclusion

Private-information risk is a determinant of asset returns. The market compensates investors for holding stocks with higher probabilities of informed trading.

Table VIII
Alternative Explanations

The table contains time-series averages of the coefficients in cross-sectional asset-pricing tests using standard Fama and MacBeth (1973) methodology and Litzenberger and Ramaswamy (L-R, 1979) precision-weighted means. The dependent variable is the percentage monthly return in excess of the one-month T-bill rate. Betas are portfolio betas calculated from the full period using 40 portfolios. *PIN* is the probability of information-based trading in stock *i* of year *t* - 1. *SIZE* is the logarithm of market value of equity (*ME*) in firm *i* at the end of year *t* - 1, and *BM* is the logarithm of the ratio of book value of common equity to market value of equity for firm *i* in year *t* - 1. *SPREAD* is the average daily opening percentage spread for firm *i* in year *t* - 1. *STD* is the standard deviation of daily returns for firm *i* in year *t* - 1. *TURNOVER* is the logarithm of the average monthly turnover year *t* - 3 to *t* - 1, and *CVTURN* is the logarithm of the coefficient of variation of the monthly turnover year *t* - 3 to *t* - 1. *T*-values are given in parentheses.

	<i>Beta</i>	<i>PIN</i>	<i>SIZE</i>	<i>BM</i>	<i>SPREAD</i>	<i>STD</i>	<i>TURNOVER</i>	<i>CVTURN</i>
Fama-MacBeth	-0.146 (-0.407)		0.106 (2.122)	0.037 (0.353)	-0.060 (-0.745)			
L-R WLS	-0.425 (-5.47)		0.081 (4.333)	0.035 (0.930)	-0.060 (-2.66)			
Fama-MacBeth	-0.164 (-0.458)	1.694 (2.483)	0.144 (2.849)	0.044 (0.419)	-0.052 (-0.650)			
L-R WLS	-0.441 (-5.66)	1.866 (3.882)	0.124 (5.746)	0.041 (1.100)	-0.051 (-2.27)			
Fama-MacBeth	0.159 (0.486)		0.053 (1.138)	-0.009 (-0.088)		-0.318 (-2.73)		
L-R WLS	0.008 (0.102)		0.034 (2.172)	-0.016 (-0.416)		-0.354 (-13.0)		
Fama-MacBeth	0.141 (0.435)	1.411 (2.104)	0.084 (1.748)	-0.002 (-0.024)		-0.311 (-2.67)		
L-R WLS	-0.008 (-0.100)	1.405 (2.932)	0.064 (3.458)	-0.010 (-0.271)		-0.345 (-12.6)		
Fama-MacBeth	0.047 (0.144)		0.119 (2.166)	0.051 (0.486)		-0.305 (-3.27)	-0.210 (-2.44)	
L-R WLS	-0.237 (-2.23)		0.134 (2.538)	0.046 (0.388)		-0.336 (-3.25)	-0.178 (-0.55)	

Table VIII: Alternative Explanations

Read: Table VIII adds alternative explanatory variables: spread, return volatility, turnover, and turnover volatility. *PIN* remains positive and significant in the joint specification.

Economic message: The *PIN* premium is not absorbed by standard liquidity, volatility, or volume channels.

13.2 Economic intuition

Informed trading creates an adverse-selection risk that cannot be fully diversified away by uninformed investors. If investors do not know which stocks informed traders favor or avoid, they cannot simply construct a portfolio that eliminates this disadvantage.

13.3 Incremental contribution revisited

The paper’s main contribution is to connect microstructure and asset pricing using a structural, stock-level information-risk measure. It provides evidence that the trading process itself contains priced information about expected returns.

13.4 Limitations

The paper relies on a specific sequential-trade model and annual estimates of PIN . The model may omit some forms of information, institutional trading, or dynamic strategic behavior. The sample is NYSE-only and covers 1983-1998, so generalizing to other markets, modern electronic trading, or post-decimalization data requires additional work.

13.5 Takeaway

The most important lesson is that price efficiency is a process, not just an end state. How prices become informative affects the risk of holding stocks, and this information risk is priced.